

Preconditioning the Regression of Power Amplifier Behavioral Models and Digital Predistorters

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Abstract—This work presents insights into the regression of power amplifier behavioral models and digital predistorters. First, the least squares regression equation is split into its basic terms and these are analyzed from a correlation approach. Techniques such as normalization, whitening, coefficient selection, coefficient exclusion and Ridge regularization and their effect on the condition number are discussed. Experimental results highlight how these techniques affect the performance of the predistortion of a class J PA working in its saturation regime in a dual band scenario, concluding that the presented methods are beneficial in the regression.

I. INTRODUCTION

The high volume of coefficients in Volterra-based models is problematic both from numerical and computational points of view. On the one hand, a large number of regressors generated by combinations of the input signals produces high correlations in matrices that will be inverted. The high correlation is translated into ill conditioning, thus producing solutions that exhibit a high dependence of the noise and small changes in the input signal. On the other hand, the complexity due to a high number of components may exceed the computing capabilities of the digital predistorter (DPD) device or lead to energy misuse in computations that may be superfluous. Selection techniques aim to identify the most relevant coefficients of a model, and are often presented as closed algorithms with different features, being their differences very subtle. Classical techniques, such as matching pursuit (MP), orthogonal MP (OMP), doubly OMP (DOMP) [1], sparse Bayesian learning (SBL) [2], dynamic model sizing (DMS) [3] and subspace pursuit (SP) are based on the same building blocks and their internal operation is affectedly inspired by common principles.

In this work we present insights on operations that enhance the regression of power amplifier (PA) behavioral models and DPD, as well as an analysis on the basic blocks that construct selection techniques. The experimental part deals with the linearization of a class J PA working in a dual-band scenario and showcases the previously discussed enhancements.

The rest of this work is organized as follows. First, Section II reviews the model regression and proposes enhancements in Volterra-based models estimation. Next, Section III covers the experimental design and the results are shown in Section IV. Section V wraps up the work with the conclusions.

II. FRAMEWORK

The measurement equation represents the relation between the complex envelope input of the signal $\mathbf{x}$ to its output $\mathbf{y}$ with

$$\mathbf{y} = \mathbf{X}\mathbf{h} + \mathbf{e}, \quad (1)$$

where $\mathbf{y} \in \mathbb{C}^{m \times 1}$ is a set output samples, $\mathbf{h} \in \mathbb{C}^{n \times 1}$ is the vector holding the model coefficients, $\mathbf{e}$ accounts for the model errors, and $\mathbf{X} \in \mathbb{C}^{m \times n}$ is a centered measurement matrix that holds one Volterra regressor in each of its columns, which are shaped by the model structure and generally consist on products of delayed and/or conjugate versions of the input signal $\mathbf{x}$. The estimation of the Volterra kernel vector $\hat{\mathbf{h}}$ through a least squares (LS) regression leads to the relation

$$\hat{\mathbf{h}} = (\mathbf{X}^H \mathbf{X})^{-1} \mathbf{X}^H \mathbf{y} = \mathbf{X}^\dagger \mathbf{y}, \quad (2)$$

where the output signal $\mathbf{y}$ is projected over the space represented by $\mathbf{X}$ through the Moore-Penrose pseudoinverse matrix $\mathbf{X}^\dagger$, that is known to bring instability to the regression because of numerical issues due to ill-conditioned Volterra matrices.

Internally, the regression is dominated by the statistics and correlations of the signals. Defining the autocovariance matrix $\mathbf{R}_\mathbf{X}$, that accounts for the relations between the basis functions, as

$$\mathbf{R}_\mathbf{X} = \mathbf{X}^H \mathbf{X}, \quad (3)$$

the precision matrix $\mathbf{P}_\mathbf{X}$ as its inverse, i.e., $\mathbf{P}_\mathbf{X} = \mathbf{R}_\mathbf{X}^{-1}$, and the cross covariance matrix $\mathbf{R}_{\mathbf{X}\mathbf{y}}$ as

$$\mathbf{R}_{\mathbf{X}\mathbf{y}} = \mathbf{X}^H \mathbf{y}, \quad (4)$$

and substituting in (2), the regression becomes

$$\hat{\mathbf{h}} = \mathbf{R}_\mathbf{X}^{-1} \mathbf{R}_{\mathbf{X}\mathbf{y}} = \mathbf{P}_\mathbf{X} \mathbf{R}_{\mathbf{X}\mathbf{y}}. \quad (5)$$

This estimation is performed in two internal steps. First, the projection of the output signal in the model basis functions, represented by $\mathbf{R}_{\mathbf{X}\mathbf{y}}$, defines the value of the coefficient. Note that this projection measures the similarity between each basis function and the output no matter how correlated the regressors are. Next, the autocovariance matrix is inverted to account for the relations between the regressors and the differences in their norm. The precision matrix is then applied to the projection to decorrelate the coefficients.

There exist some approaches to diminish the inconveniences of ill conditioning in the autocovariance matrix inversion and the computational complexity of this mathematical operation:

1) *Normalization*: Let $\mathcal{N}$ be a normalization matrix made up of the ℓ_2 -norm of the regressors:

$$\mathcal{N} = \text{Dg} \left(\sqrt{\mathbf{R}_{\mathbf{X}}} \right) \quad (6)$$

where $\text{Dg}(\cdot)$ returns a diagonal matrix whose entries are the elements in the diagonal of its argument. Defining the normalized regressors matrix as $\bar{\mathbf{X}} = \mathbf{X} \cdot \mathcal{N}^{-1}$ and the normalized Volterra kernel vector $\bar{\mathbf{h}} = \mathcal{N} \cdot \mathbf{h}$, results in a regression that follows

$$\hat{\mathbf{h}} = \mathcal{N}^{-1} (\bar{\mathbf{X}}^H \bar{\mathbf{X}})^{-1} \bar{\mathbf{X}}^H \mathbf{y}. \quad (7)$$

Generally, the normalization process produces a lower condition number in $\mathbf{R}_{\bar{\mathbf{X}}}$ than in $\mathbf{R}_{\mathbf{X}}$.

2) *Whitening*: A different procedure consists in defining a transformation matrix $\mathcal{T}$ that relates the Volterra space $\mathbf{X}$ and the orthonormal axes $\mathbf{Z}$:

$$\mathbf{Z} = \mathbf{X} \mathcal{T}. \quad (8)$$

Such transformation is invertible through $\mathbf{X} = \mathbf{Z} \mathcal{T}^{-1}$. If the estimation is performed over the orthogonal basis, the regression results in

$$\hat{\mathbf{h}} = \mathcal{T} \mathbf{R}_{\mathbf{Z}\mathbf{y}} = \mathcal{T} \mathbf{Z}^H \mathbf{y}, \quad (9)$$

in which the calculation of the covariance matrix inverse has disappeared, and as its result the regression consists in just calculating the projection matrix $\mathbf{R}_{\mathbf{Z}\mathbf{y}}$ and undo the transformation $\mathcal{T}$ afterwards. This approach omits the inversion of the covariance matrix at the expense of the quality of the selected transformation. Since this transformation depends on the statistics on the signal and not on the specific realization, its computation can be performed only once, reducing run time.

3) *Coefficient selection*: Coefficient selection techniques aim at producing the set of regressor indices S , known as the support set, that identify the most relevant basis functions of the model. The family of techniques that perform coefficient selection independently of the model structure, i.e. greedy pursuits, generally build decisions on the correlation matrices $\mathbf{R}_{\mathbf{X}}$ and $\mathbf{R}_{\mathbf{X}\mathbf{y}}$. The base of model pruning is keeping the coefficients that exhibit a larger value in their Volterra coefficient h_i , and the main difference between the techniques is the way they deal with the correlation between the components. The key hard decision of the techniques are:

- MP: chooses the component that has the maximum value of $\mathbf{R}_{\mathbf{X}\mathbf{y}}$.
- DMS (selection part): selects the component that has the maximum value in $\mathbf{R}_{\mathbf{X}\mathbf{y}}$ normalized by their standard deviation, i.e., the diagonal elements of $\mathbf{P}_{\mathbf{X}}$.
- OMP: selects the component that has the maximum value of $\mathbf{R}_{\mathbf{X}\mathbf{y}}$ after subtracting the effect of the previously selected components.
- DOMP: selects the component that has the maximum value of $\mathbf{R}_{\mathbf{X}\mathbf{y}}$ after orthogonalizing the residual error and the rest of the basis functions with respect the selected regressor.

- SBL: maximizes a logarithmic function of a combination of $\mathbf{R}_{\mathbf{X}}$ and $\mathbf{R}_{\mathbf{X}\mathbf{y}}$.
- SP: selects the highest absolute value of the coefficient in the set of basis functions with the highest correlation with the residual.

All the selection techniques converge to selecting the coefficients that exhibit the maximum value of $\mathbf{R}_{\mathbf{X}\mathbf{y}}$ when working with orthonormal basis functions. Let $\mathbf{X}_S$ be the matrix whose columns are those indicated by S . The regression in this case is attained through

$$\hat{\mathbf{h}} = \mathbf{R}_{\mathbf{X}_S}^{-1} \mathbf{R}_{\mathbf{X}_S \mathbf{y}} = \mathbf{P}_{\mathbf{X}_S} \mathbf{R}_{\mathbf{X}_S \mathbf{y}}. \quad (10)$$

4) *Coefficient exclusion*: In the pool of regressors there might be coefficients with such a low impact in the output signal that can be omitted in the selection process. Those elements that show a cross correlation lower than a threshold can be dropped from the list of potential regressors. This threshold can be related to the signal-to-noise ratio, and it can also be used as a stopping criteria: once the residual reaches the threshold value, there is no need for analyzing more components. In the case of using orthogonalization, coefficient exclusion can be applied after decorrelation, since this operation decreases the value of the attained component.

5) *Ridge regularization*: Ridge regularization adds a constraint on the cost function to limit the absolute value of the coefficients. Generally speaking, this type of regression trades bias and variance of the estimator, and it takes the form of

$$\hat{\mathbf{h}} = (\mathbf{R}_{\mathbf{X}} + \lambda \mathbf{I})^{-1} \mathbf{R}_{\mathbf{X}\mathbf{y}}, \quad (11)$$

where λ is the shrinkage parameter. The regularization effect is to reduce the variance of the estimator as λ grows.

III. EXPERIMENTAL DESIGN

In order to highlight the above-mentioned features, a direct learning architecture (DLA) DPD was implemented with a concurrent dual-band input signal. The experimental test bench was composed of an SMU200A vector signal generator (VSG) from Rohde & Schwarz, a PXA-N9030A vector signal analyzer (VSA) from Keysight Technologies and a dc power supply. The RF output port of the VSG was fed to a Mini-Circuits TVA-4W-422A+ preamplifier and then applied to the device under test. The PA to be linearized was a continuous mode class J amplifier based on the CGH35030F GaN HEMT from Cree for the frequency range 700-1000 MHz. The operating point was set to an average output power of +24 dBm.

The dual-band probe signal was composed by two orthogonal frequency division multiplexing (OFDM) signals according to the fifth generation New Radio (5G-NR) standard, each with 10 MHz bandwidth, 30 kHz subcarrier spacing and peak-to-average power ratio (PAPR) about 11 dB. Before combining both baseband signals, generated with a sample frequency $f_s = 92.16$ MHz, frequency offsets $\pm f_s/4$ were respectively applied to each band. These values were chosen to optimize the observable frequency range. The dual-band waveform was employed to modulate a 800 MHz carrier. The same sampling frequency f_s was employed in the VSA. The input and output

TABLE I
LINEARIZATION RESULTS OF THE TECHNIQUES IN COMPARISON IN THE 20TH DIRECT LEARNING ARCHITECTURE ITERATION

Case		NMSE _L	NMSE _R	ACPR _{-1L/+1L/-1R/+1R}	EVM _L	EVM _R	Condition number (dB)	Normalized run time
		(dB)		(dBc)	(%)			
w/o DPD	-	-27.3	-27.9	-42.9 / -44.1 / -43.7 / -42.9	4.3	4.1	-	-
LS	(2)	Divergence						779
LS + norm	(7)	Divergence						185
LS + DOMP	(10)	-48.8	-48.5	-60.1 / -61.3 / -62.6 / -58.1	1.2	1.2	431	1.00
LS + norm + DOMP	(7)+(10)	-48.8	-48.3	-60.3 / -61.5 / -62.7 / -58.2	1.2	1.2	86	1.02
Ridge + norm	(7)+(11)	-48.3	-48.1	-59.8 / -61.2 / -63.8 / -59.5	1.2	1.2	72	17.60
Ridge + DOMP + norm	(11)+(7)+(10)	-48.8	-48.4	-60.1 / -61.4 / -62.5 / -58.1	1.2	1.2	63	1.02

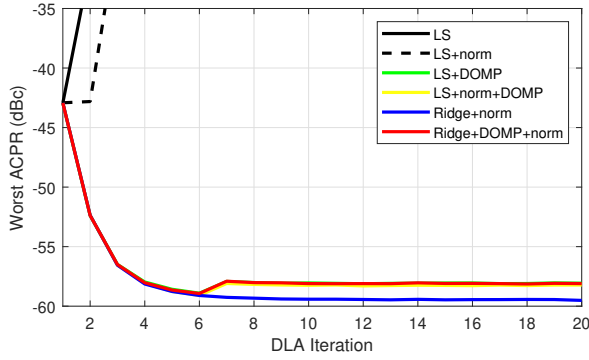


Fig. 1. Worst adjacent channel power ratio (ACPR), i.e., highest ACPR of the four possible values, versus iteration number for the techniques in comparison.

signals were split into two single-band signals with a sampling frequency $f_s/2$, and then used to compute the DPDs.

The 2D-GMP model [4] was selected to perform the regression with high nonlinear order and memory length to compensate the asymmetry and memory effects exhibited by the class J PA, thus yielding 534 regressors. It is worth mentioning that this model is prone to numerical errors due to its large amount of regressors, being mandatory the use of enhancement techniques to avoid divergence of the DLA-DPD.

The DLA was executed for 20 iterations mixing different combinations of LS (2), norm (7), DOMP (10) and Ridge (11). Coefficient selection was performed through DOMP as a reference, reducing the models to 49 and 51 coefficients for each band, and the Ridge shrinkage parameter was obtained through cross validation.

IV. RESULTS

In this section, the proposed techniques are benchmarked regarding their linearization capabilities and numerical characteristics in the regression. Fig. 1 shows the linearization adjacent channel power ratio (ACPR) in the worst-case side channel versus the number of DLA iterations for all the techniques in comparison. In this case, all the techniques perform similarly except LS and LS+norm, that diverge in the DLA linearization process due to the high model complexity.

Table I presents the linearization metrics of normalized mean squared error (NMSE), ACPR, and error vector magni-

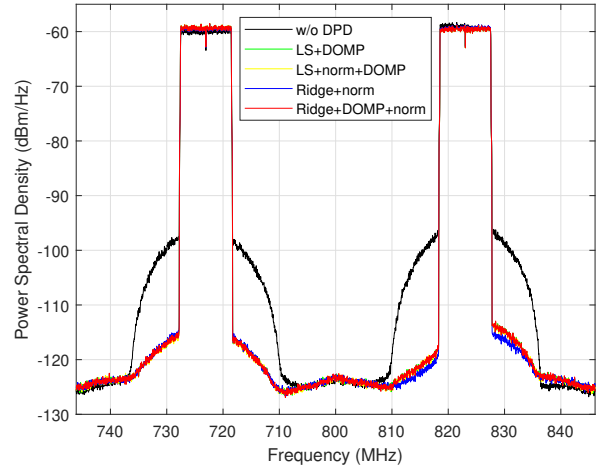


Fig. 2. Power spectral densities (PSDs) of the output signal without predistorter and with predistorter attained with the techniques under comparison.

tude (EVM) for the benchmark. The techniques that succeed in the predistortion of the PA achieve similar results. The condition number of the inverted matrix, defined as the ratio of its largest singular value to the smallest, in the regression decreases with normalization, coefficient selection and regularization techniques. Normalized run time expresses the execution time of the regression, omitting pruning and cross validation time. This parameter is primarily affected by the application of a coefficient selection technique.

Finally, Fig. 2 presents the power spectral densities (PSDs) of the output signal without DPD and with DPD. It can be observed that the techniques achieve to reduce spectral regrowth from -43 dBc to less than -58 dBc in all the cases.

V. CONCLUSION

In this work, several techniques for enhancing the regression of PA and DPD Volterra models have been presented. Experimental results showed that although the linearization performance is equivalent, the way the regression is performed affects to its conditioning and the run time, concluding that normalization, coefficient selection and regularization are beneficial to the regression.

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